

ACADEMIC PERFORMANCE MONITORING SYSTEM

TEAM MEMBERS :

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Aim:

To analyze and visualize student academic performance by calculating total marks, average marks, attendance, and categorizing the student performance using R.

Objectives:

- 1.To create and manage student academic data using R data frames.
- 2.To calculate total and average marks for each student.
- 3.To categorize students based on their performance level.
- 4.To determine pass or fail status based on their average marks.
- 5.To visualize student performance using graphical methods.
- 6.To analyze the relationship between attendance and academic performance.
- 7.To provide a simple and effective method for monitoring student progress.

Software requirements:

1.R Programming Language

- It is used as the core technology of the project and performs data analysis, calculations, and statistical operations.

2.RStudio

- Integrated Development Environment (IDE) for R.It provides code editor, console, and plotting panel.

3.Libraries: ggplot2, dplyr

- ggplot2 is used for data visualization.It helps to create bar charts and scatter plots.
- dplyr is used for data manipulation.It helps in creating new variables and modifying datasets.

Methodology:

The project follows these steps:

1. Load required R libraries
2. Create a student dataset
3. Calculate total and average marks
4. Classify performance levels
5. Identify pass/fail status
6. Display the dataset
7. Visualize performance using charts

Procedure:

Step 1: Create a new R Script file and save the file as student_performance.R

Step 2: Load the required R library ggplot2 to create the graphs.

Step 3: Load the required R library dplyr for data manipulation.

Step 4: Create a data frame containing student details such as,

- Student ID
- Student Name
- Marks in three subjects
- Attendance percentage

Step 5: Calculate the Total Marks by adding Marks1, Marks2, and Marks3 of each student.

Step 6: Calculate the Average Marks for each student by dividing total marks by 3.

Step 7: Categorize student performance based on average marks.

- If students score an average ≥ 80 marked as Excellent.
- If students score average ≥ 60 and < 80 marked as Average.
- Others marked as Poor (Average < 60)

Step 8: Determine the academic result as Pass or Fail based on average marks.

- Students with average ≥ 50 are marked as Pass
- Others are marked as Fail

Step 9: Display the student data using the print() function.

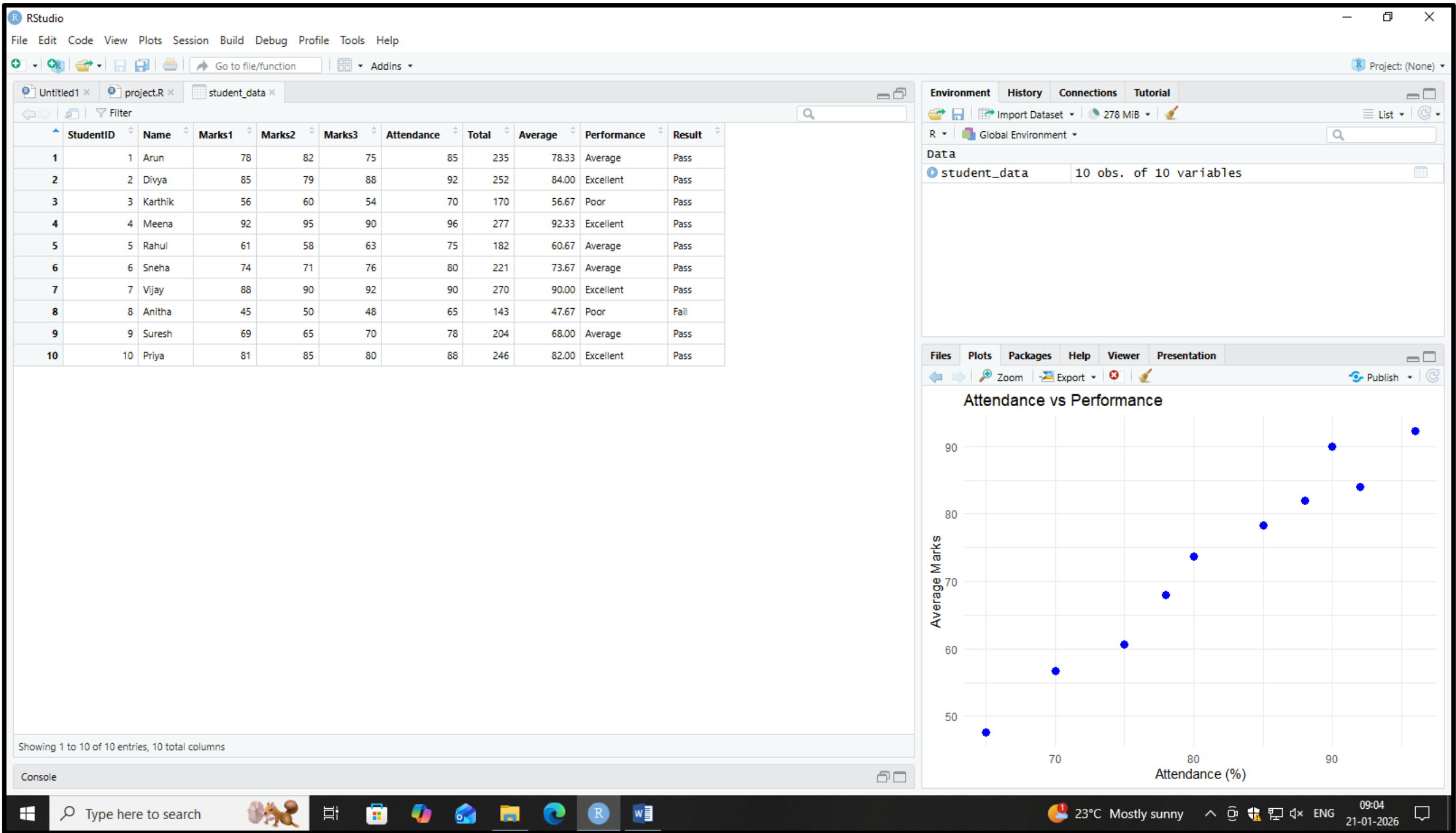
Step 10: Create a bar chart to show average marks of students based on performance category.

Step 11: Create a scatter plot to show the relationship between attendance and average marks.

Dataset:

The dataset consists of 10 students with the following attributes:

Attribute	Description
StudentID	Unique student identification number
Name	Student name
Marks1	Marks in Subject 1
Marks 2	Marks in Subject 2
Marks3	Marks in Subject 3
Attendance	Attendance percentage
Total	Total marks
Average	Average marks
Performance	Performance category
Result	Pass/Fail status



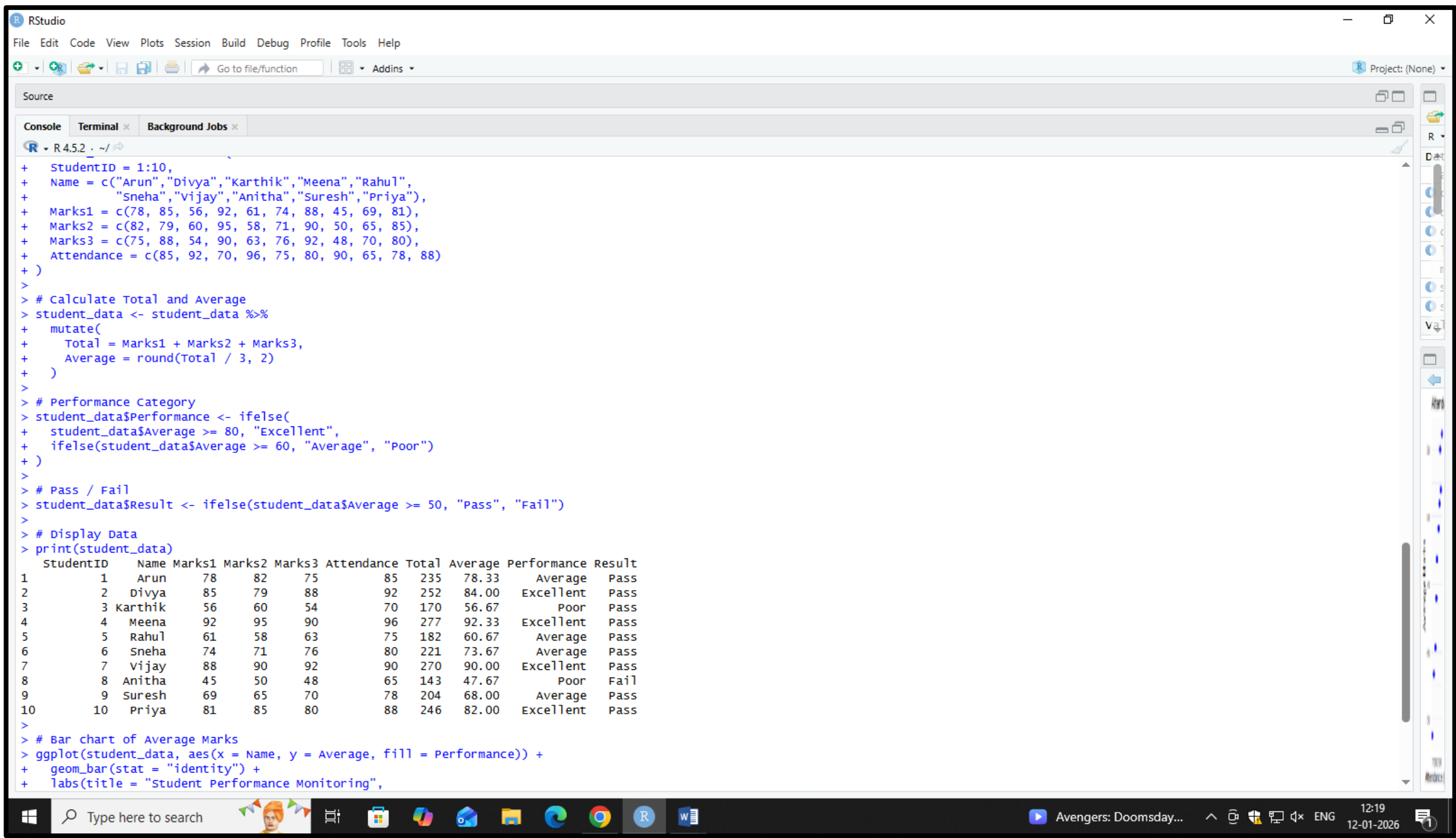
Program:

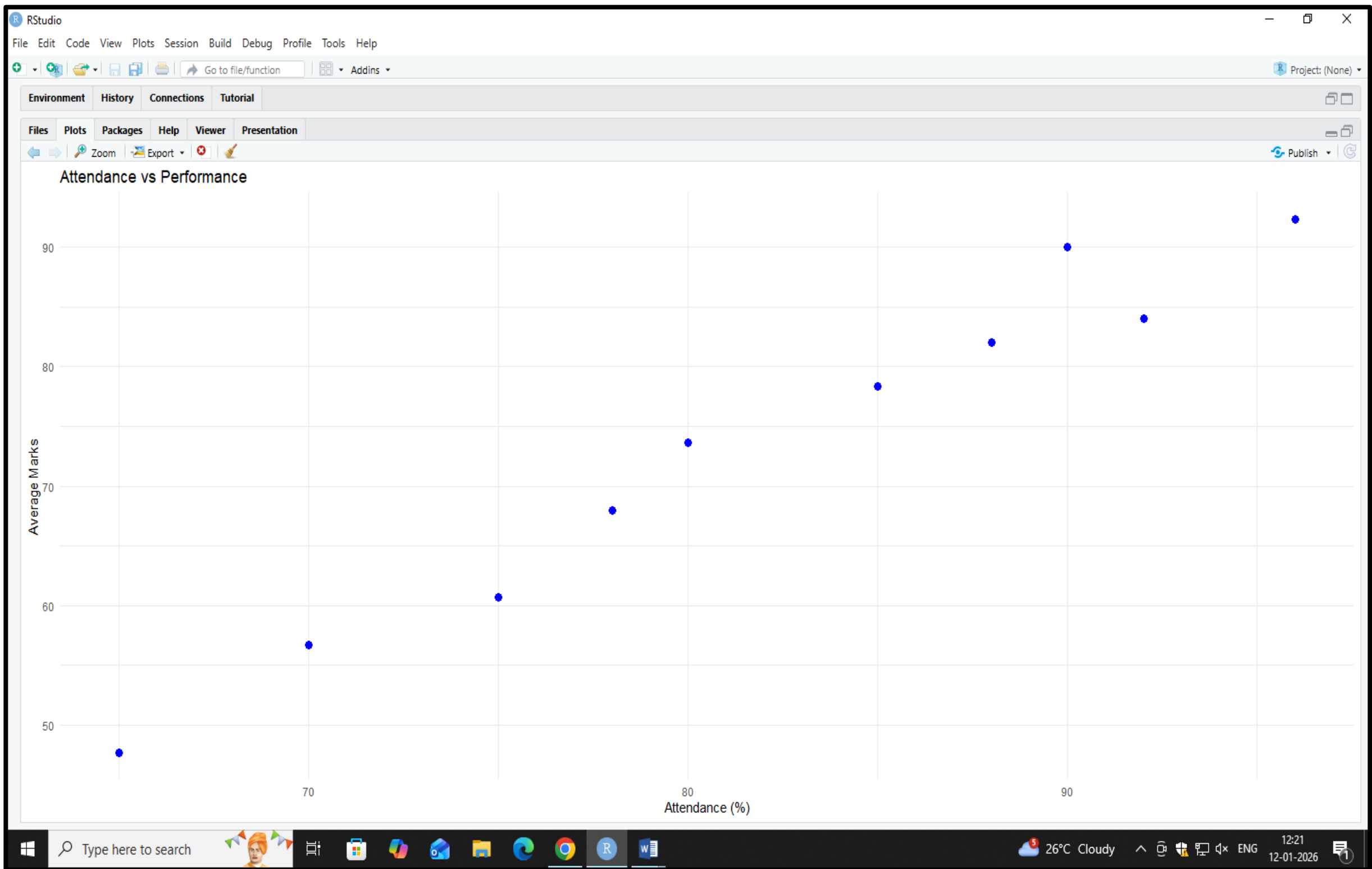
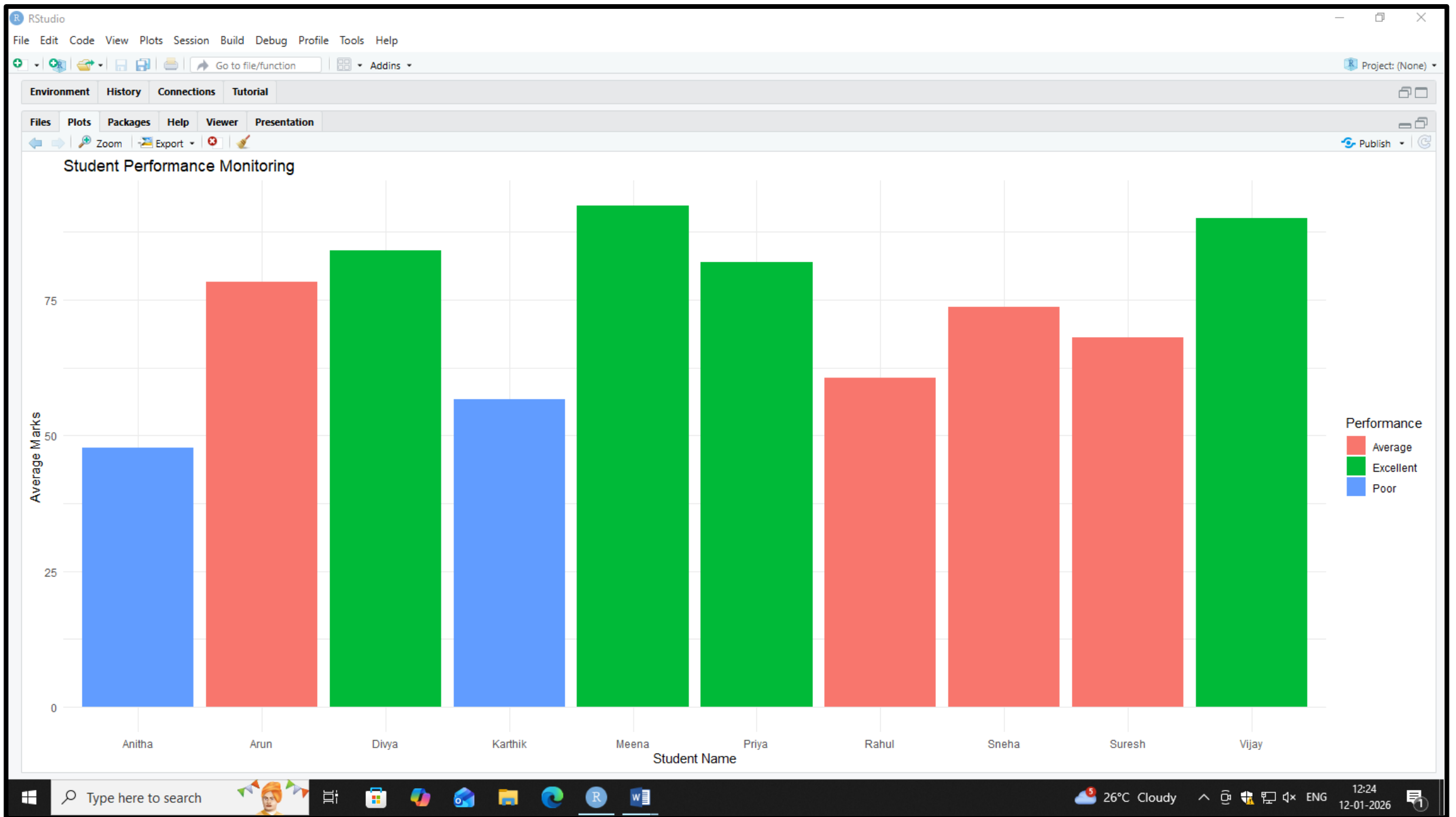
```
# Load required libraries
install.packages( "ggplot2" )
library(ggplot2)
install.packages( "dplyr" )
library(dplyr)
# Create student dataset
student_data <- data.frame(
  StudentID = 1:10,
  Name = c("Arun","Divya","Karthik","Meena","Rahul",
  "Sneha","Vijay","Anitha","Suresh","Priya"),
  Marks1 = c(78, 85, 56, 92, 61, 74, 88, 45, 69, 81),
  Marks2 = c(82, 79, 60, 95, 58, 71, 90, 50, 65, 85),
  Marks3 = c(75, 88, 54, 90, 63, 76, 92, 48, 70, 80),
  Attendance = c(85, 92, 70, 96, 75, 80, 90, 65, 78, 88)
)
# Calculate Total and Average
student_data <- student_data %>%
mutate(
  Total = Marks1 + Marks2 + Marks3,
  Average = round(Total / 3, 2)
)
# Performance Category
student_data$Performance <- ifelse(
  student_data$Average >= 80, "Excellent",
  ifelse(student_data$Average >= 60, "Average", "Poor")
)
# Pass / Fail
student_data$Result <- ifelse(student_data$Average >= 50, "Pass", "Fail")
# Display Data
print(student_data)

# Bar chart of Average Marks
ggplot(student_data, aes(x = Name, y = Average, fill = Performance)) +
  geom_bar(stat = "identity") +
  labs(title = "Student Performance Monitoring",
  x = "Student Name",
  y = "Average Marks") +
  theme_minimal()
# Attendance vs Average Marks
ggplot(student_data, aes(x = Attendance, y = Average)) +
```

```
geom_point(color = "blue", size = 3) +
labs(title = "Attendance vs Performance",
x = "Attendance (%)",
y = "Average Marks") +
theme_minimal()
```

Output:





Project Report:

ACADEMIC PERFORMANCE MONITORING SYSTEM

This project analyzes student academic performance using R programming. Student marks and attendance data are processed to calculate total and average marks. Based on the average, students are categorized as Excellent, Average, or Poor, and their pass/fail status is determined.

The project uses ggplot2 and dplyr libraries for data manipulation and visualization. Bar charts display student performance, while scatter plots show the relationship between attendance and average marks. This system helps educators easily monitor academic performance and identify students who need improvement.

Result:

Thus, the program effectively demonstrates student performance monitoring using R programming.

- Total and average marks were calculated for all students.
- Students were classified into Excellent, Average, and Poor performance categories.
- Pass and Fail status was determined accurately.
- Graphs clearly represented student performance and attendance impact.

Hence, the program was executed successfully.